

ADEN INTL, Yemen

WMO: 414800

Lat: 12.833N

Lon: 45.033E

Elev: 10

StdP: 14.69

Time Zone: 3.00 (E03)

Period: 02-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	70.1	71.7	52.2	57.9	82.5	56.3	67.5	81.1	24.4	79.6	23.0	79.3	4.9	30	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.0	98.6	76.7	97.1	77.1	96.5	77.2	83.9	91.8	83.0	91.2	82.3	90.7	11.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.1	167.3	89.8	80.8	160.0	89.6	80.2	156.6	89.4	48.2	91.9	47.1	91.6	46.3	90.1	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.1	21.2	19.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	56.9	85.5	2.9	1.7	54.8	86.7	53.1	87.7	51.5	88.6	49.4	89.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.6	78.4	78.4	79.9	83.9	88.3	91.6	90.7	89.9	89.6	84.7	81.1	78.8
	DBStd	5.35	1.34	1.59	2.43	1.84	2.31	1.66	1.64	1.73	1.92	2.42	2.10	1.92
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12643	879	796	926	1017	1187	1249	1262	1237	1189	1076	932	893
	CDD65	7168	414	376	461	567	722	799	797	772	739	611	482	428
	CDH74	91926	3102	2943	4487	7031	10360	12394	12012	11464	10891	8078	5336	3828
	CDH80	45652	255	319	1019	2833	5909	8074	7548	7007	6576	3859	1669	584
Wind	WSAvg	10.6	12.9	12.7	12.0	12.2	9.3	8.0	9.6	10.1	8.7	9.9	10.4	11.2
Precipitation	PrecAvg	2.4	0.4	0.1	0.3	0.1	0.1	0.0	0.2	0.2	0.2	0.4	0.2	0.1
	PrecMax	6.2	3.8	0.4	1.3	1.7	0.4	0.4	0.5	2.3	1.1	3.9	2.0	0.6
	PrecMin	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.1	0.0
	PrecStd	1.5	0.8	0.1	0.4	0.4	0.1	0.1	0.1	0.6	0.2	0.9	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.9	84.5	88.1	91.7	98.9	100.7	99.0	98.6	98.6	95.1	91.5	85.6
		MCWB	70.5	72.1	72.5	76.4	76.8	78.3	76.6	75.7	76.2	75.1	70.1	75.0
	2%	DB	82.7	83.3	86.3	90.0	96.7	98.7	97.2	96.9	96.9	92.7	89.2	84.4
		MCWB	71.0	72.2	74.8	77.4	79.0	77.2	76.5	75.5	77.2	76.5	73.5	73.2
	5%	DB	82.3	82.6	85.4	89.3	94.8	97.2	96.5	95.4	95.4	91.3	87.5	83.9
		MCWB	71.0	71.9	74.5	77.4	79.2	77.6	76.7	76.4	78.0	77.2	74.5	72.7
	10%	DB	81.3	82.2	84.3	87.9	93.0	96.6	95.1	94.9	94.7	89.8	86.0	82.7
		MCWB	70.7	71.6	73.6	77.2	79.6	77.8	77.3	76.6	78.3	75.9	74.1	71.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.9	75.8	78.3	80.5	84.3	86.4	83.6	82.6	84.1	81.9	78.5	77.4
		MCDB	80.9	82.7	85.2	88.0	91.2	91.1	90.9	90.3	92.3	90.1	86.3	83.6
	2%	WB	73.1	74.6	77.3	79.7	83.2	84.6	82.3	81.4	82.9	80.7	77.1	75.3
		MCDB	80.0	81.4	84.0	87.2	90.8	92.2	90.3	89.9	91.1	89.3	85.9	82.4
	5%	WB	72.5	73.7	76.6	79.1	82.3	83.4	81.1	80.6	82.3	79.8	76.1	74.1
		MCDB	79.7	80.5	83.3	86.6	90.6	91.9	90.1	89.5	90.5	88.6	85.0	81.8
	10%	WB	71.9	72.9	75.6	78.6	81.2	82.5	80.4	79.9	81.4	78.8	75.4	72.9
		MCDB	79.4	79.9	82.6	86.2	89.8	91.2	90.0	89.6	89.9	87.9	84.2	81.2
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.1	9.6	8.8	9.4	11.0	10.5	11.0	10.3	11.4	10.6	8.9
		MCDBR	7.7	8.3	9.8	9.9	12.3	12.0	11.6	11.6	11.9	12.4	12.2	9.1
		MCWBR	4.4	4.6	5.5	5.3	7.5	8.3	6.7	7.0	7.3	7.1	7.2	5.5
	5% WB	MCDBR	7.0	7.6	7.4	8.3	9.9	11.1	10.5	10.8	10.1	10.6	10.0	7.9
		MCWBR	4.5	4.6	4.3	4.6	7.0	8.3	7.1	7.0	6.9	6.7	6.4	5.2
Clear-Sky Solar Irradiance	taub	0.462	0.488	0.510	0.506	0.588	0.743	0.814	0.754	0.656	0.510	0.472	0.461	
	taud	2.177	2.089	2.044	2.083	1.881	1.569	1.481	1.582	1.735	2.047	2.152	2.177	
	Ebn at Noon	265	263	260	259	234	198	185	200	220	254	260	262	
	Edh at Noon	46	52	56	54	64	87	95	87	75	54	47	45	
All-Sky Solar Radiation	RadAvg	1694	1929	2145	2212	2201	2090	1938	2019	2030	2040	1862	1710	
	RadStd	74	80	78	79	73	81	70	55	51	57	53	79	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page